Type MS · Explain genetic and environmental causes of variation

Answer 1 of 3 · Question 3 part d · 3 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part d in this mark scheme extract

number				
3	(a)	C hypothalamus; A does not control sweating B does not control sweating D does not control sweating		1
	(b)	B motor; A transmits impulse from sensory to motor C alternative name for A with same function D transmits impulse to brain		1
	(c)	• (sweat) evaporates from (the surface of) the skin; • using heat from body/blood; • cools body/reduces body temperature;		3
	(d)	• runs in families; • must be faulty/mutated gene/chromosome/allele; • which transfers/passes on information to offspring;		3
	(e)	obesity / high sugar intake / high BMI;		1
	(f)	• exercise/activity of individual varies; • greater rate of aerobic respiration; • heat given off as by-product/increased body temperature; • external temperature varies; • hotter more sweating/cooler less sweating; • mass of individual varies; • volume of water in the body/fluid intake varies;	Allow reverse argument for mps 2 and 3	5 Total 14

Type MS · Explain genetic and environmental causes of variation

Answer 1 of 3 · Question 3 part d · 3 marks

2021 NOV · Paper 02 · extract 2 of 2 · find part d in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain genetic and environmental causes of variation

Answer 2 of 3 · Question 3 part b · 3 marks

2020 NOV · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

Number			
3 (a) (i)	• correct labelling of axes (1) • suitable scale (1) • X and Y axis correct way round (1) • plotting (2)	minus 1 for each incorrect plot	5
(ii)	Any two from • few at lower end/150-151/top end/160 (1) • majority in middle/154-157 (1) • increases to 154-155 then decreases (1)		2
(b)	Any three from • nutrition (1) • sex (1) • age (1) • genes/genetically based (1)		3

Total 10 marks

Type MS · Explain genetic and environmental causes of variation

Answer 2 of 3 · Question 3 part b · 3 marks

2020 NOV · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain genetic and environmental causes of variation

Answer 3 of 3 · Question 5 part d · 2 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part d in this mark scheme extract

5 (a)	• two monosaccharides; • joined together;		2
(b)	molecule too large;		1
(c) (i)	• transcription (of lactase gene) does not occur; • RNA is not produced/synthesised; • lactase/protein/enzyme not produced;		3
(ii)	• lactose can't be digested; • into components to be absorbed;		2
(d)	• they eat more dairy produce; • being able to digest it would give them an advantage;		2
(e)	• water passes from cells; • into gut; • diarrhoea causes poor absorption of vitamin D and Ca^{2+}; • as they are water soluble;		4

Total 14 marks

Type MS · Explain genetic and environmental causes of variation

Answer 3 of 3 · Question 5 part d · 2 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part d in this mark scheme extract

Question number	Answer	Notes	Marks